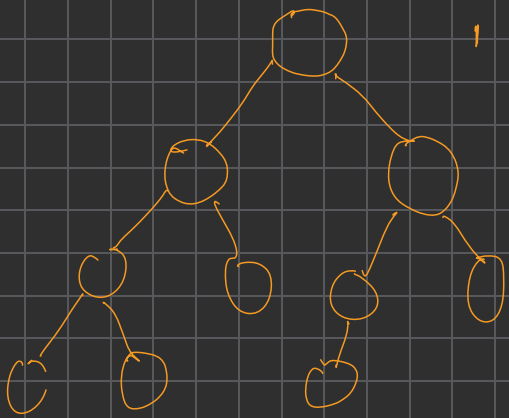
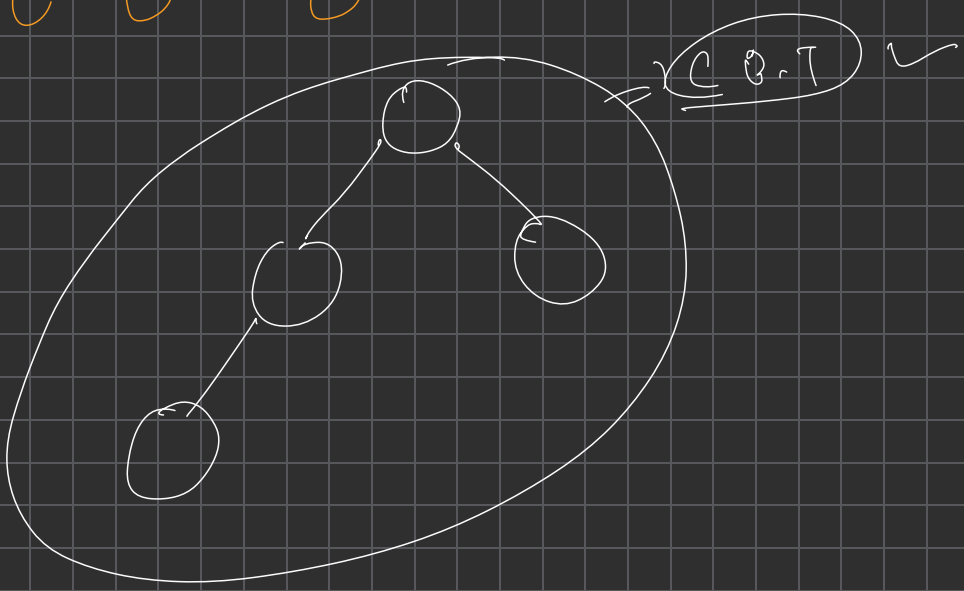
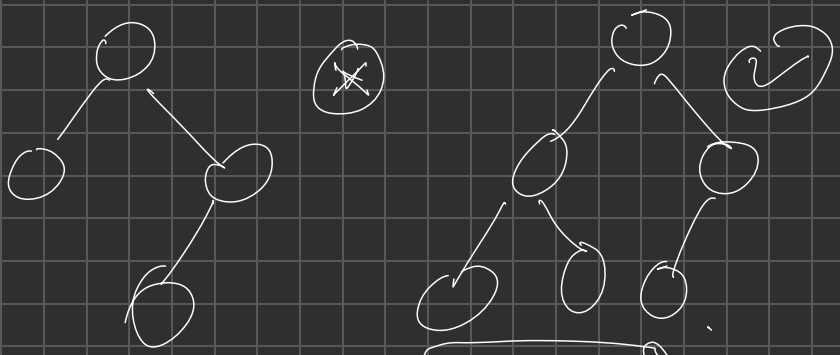
Heap : K.C ✓

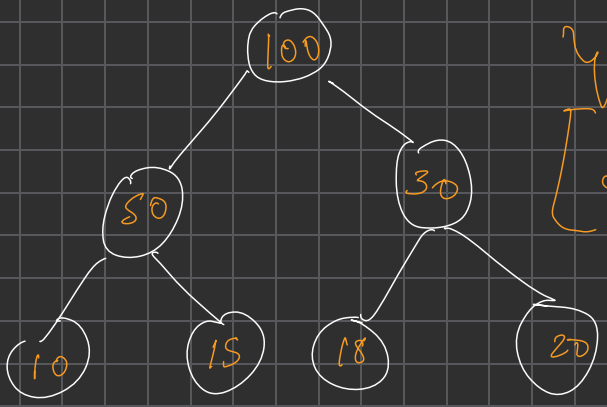
- Complete Binary Trees
- last level should be filled from left to right







Heap → Max Heap
↳ Min Heap



↳ Max heap ≠ CBT

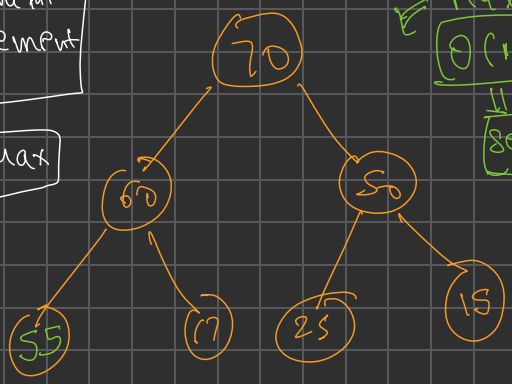
[Every parent should greater than equal to its child]

Complete Binary Tree == Height balance Binary tree

Max element root element hoga

Second Max

60



Max

O(1)

Search

Max elem

Max hip

✗ Bilkul nahi

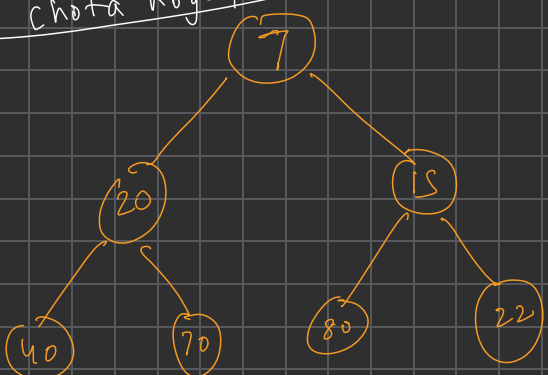
Third max

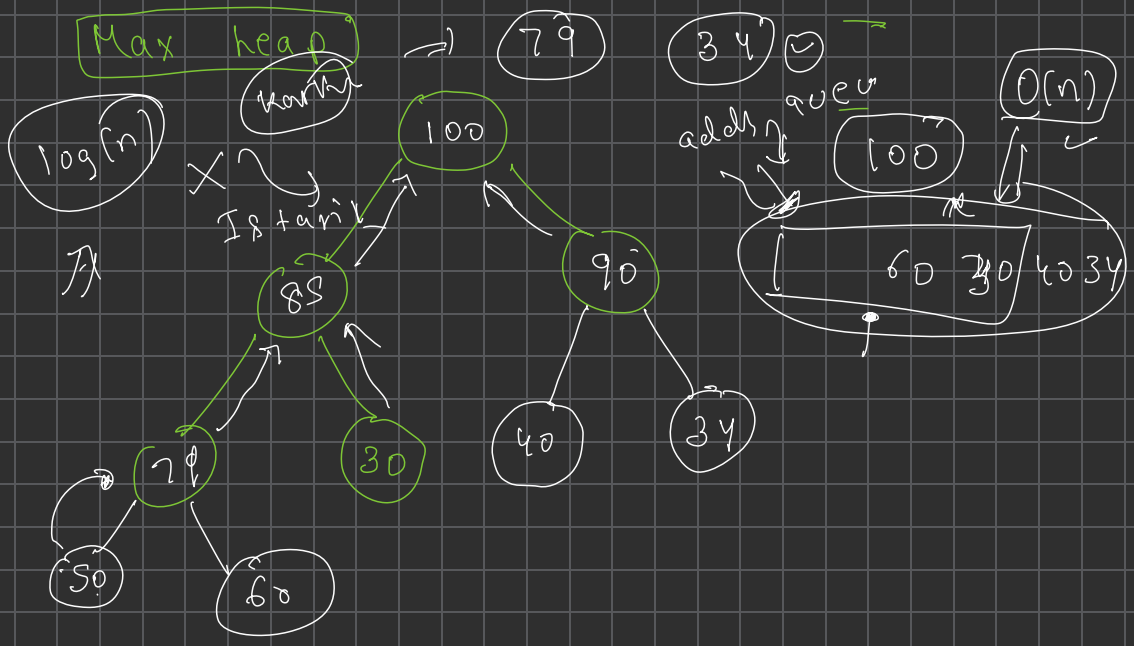
root ke 2 child

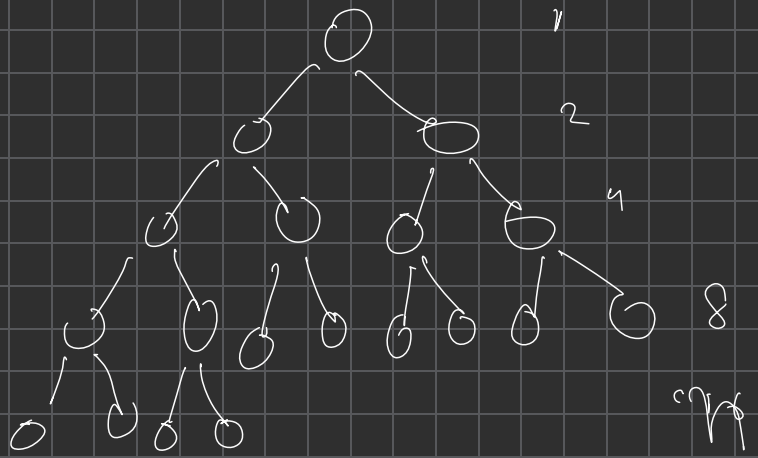
Jo chota hoga, wo mera third max ✗

Min-Heap

Parent should be less than equal to its child

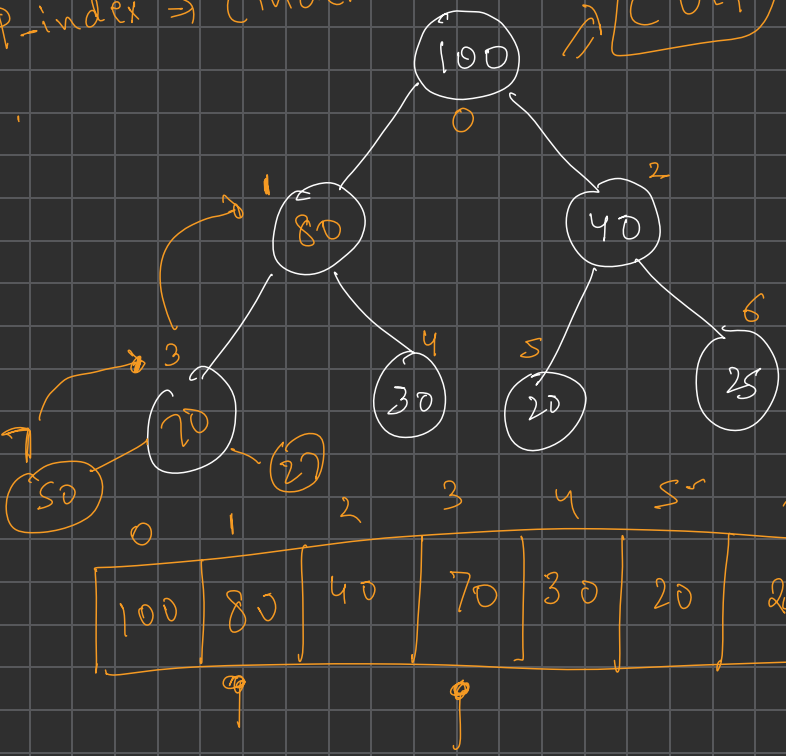



Max heap
79
34
worst
O(n)
log(n)
100
adds
queue
Is this?
85
90
70
30
40
34
50
60


1
2
4
8

Child = index;

P_index ⇒ (index-1)/2



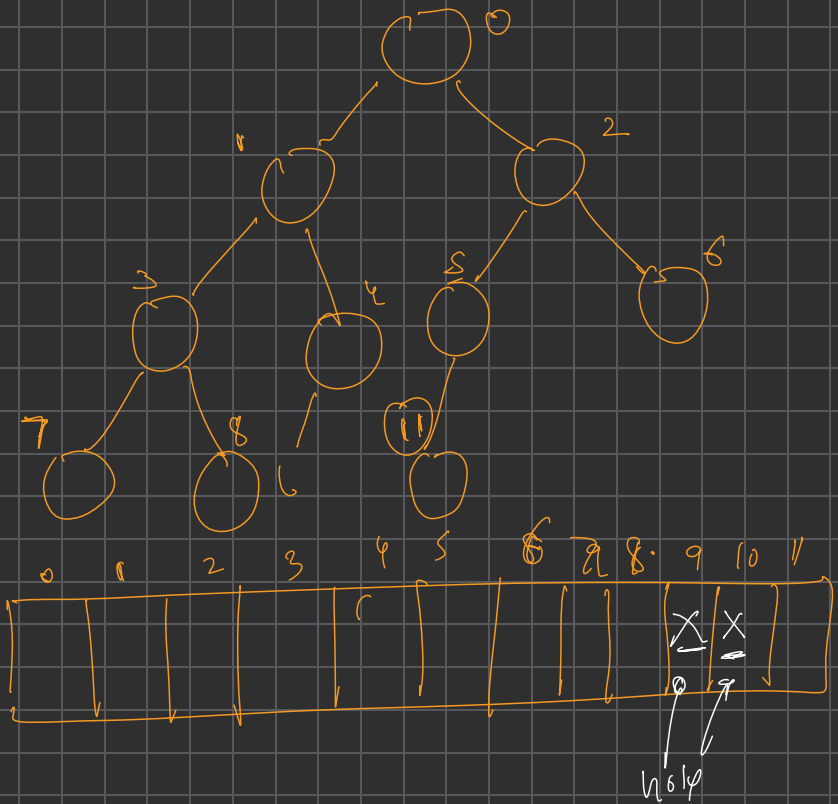
CBT ⇒ Complete Bin Tree

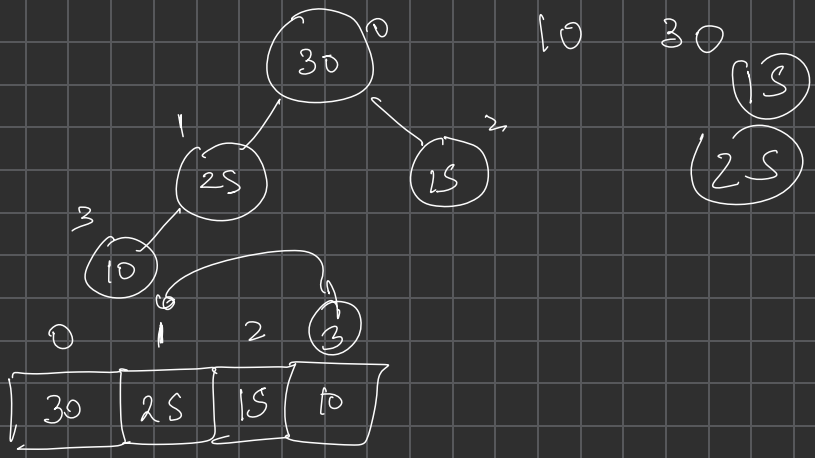
Parent = index

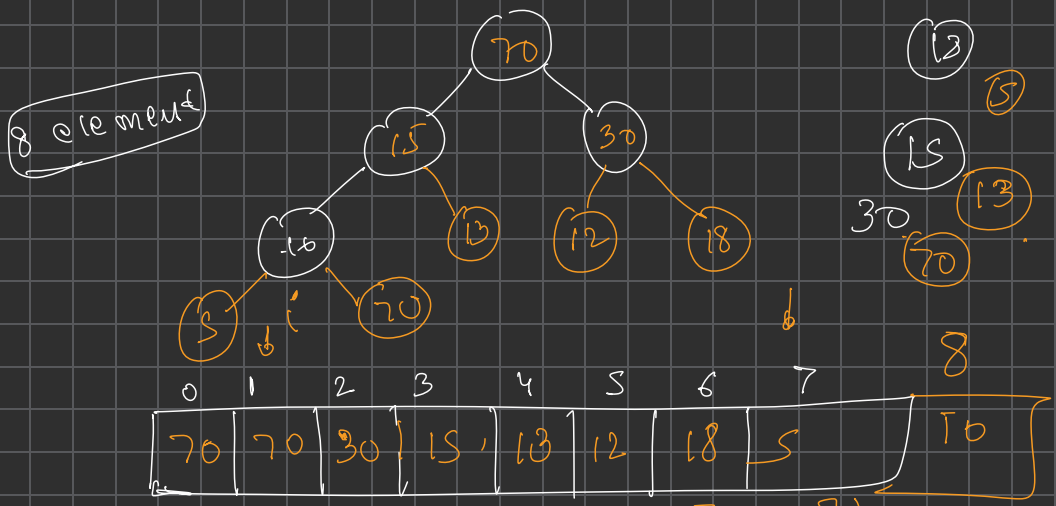
Left child ⇒ 2*index+1

Right child ⇒ 2*index+2

27




30
25
15
10
3
0
1
2
0
1
2
3
30
25
15
10
10
30
15
25



While (i ≥ 0 && arr[i] > arr[(i-1)/2])

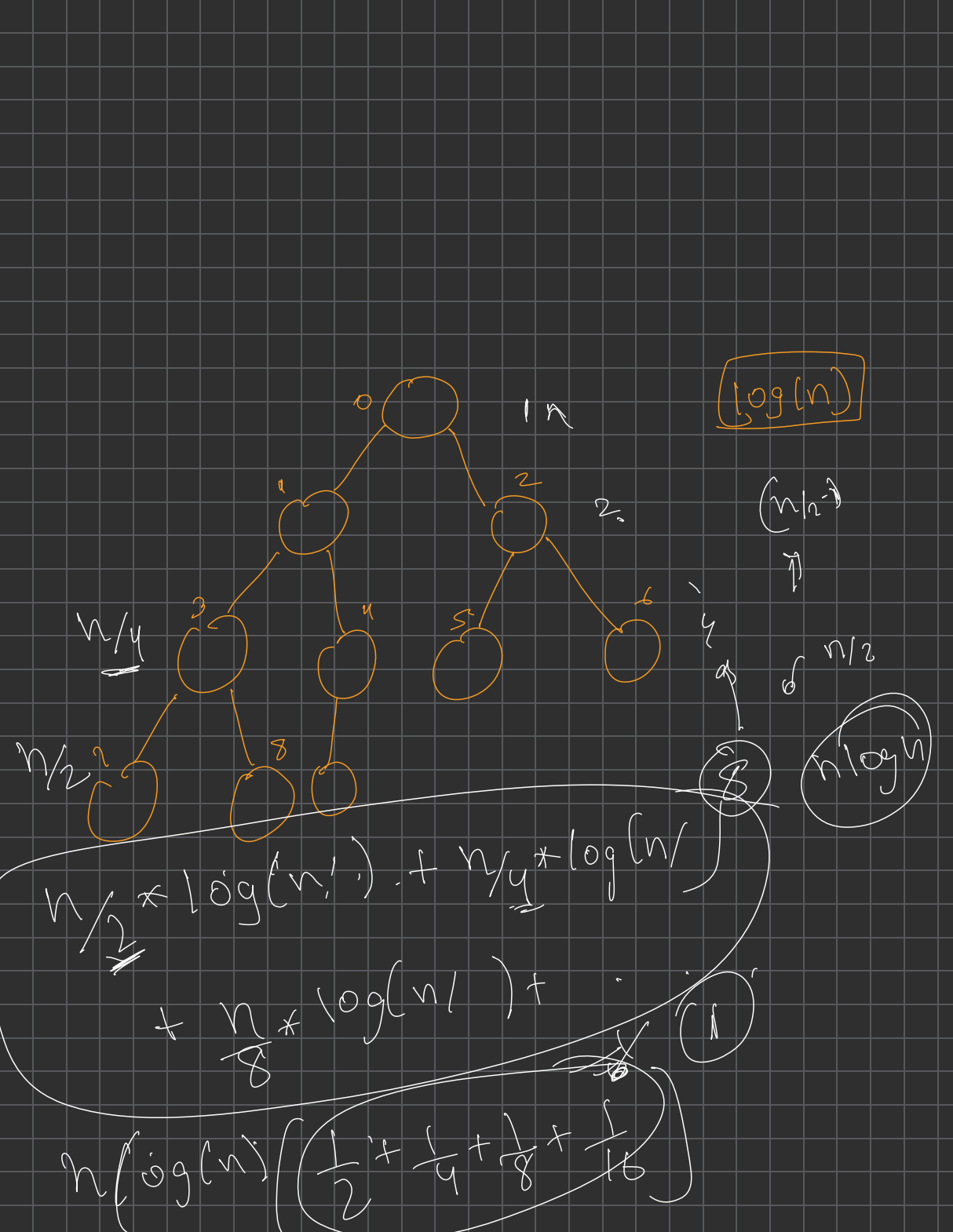
{ swap(arr[i], arr[(i-1)/2]);

i = (i-1)/2;

}

2*index +1

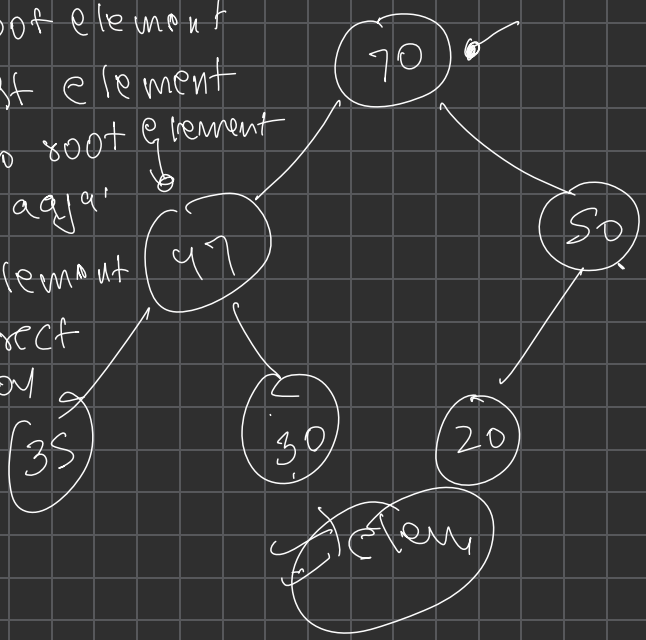
2*index +2


log(n)
n
n/2
n/4
n/8
n/16
n/2 * log(n/2) + n/4 * log(n/4) + n/8 * log(n/8) + ... (1)
n(log(n)(1/2 + 1/4 + 1/8 + 1/16

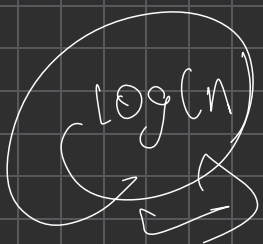
(1) Root element

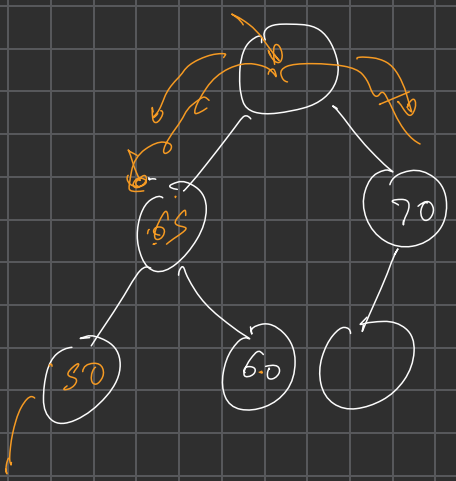
(2) Last element ko root element se replace karo

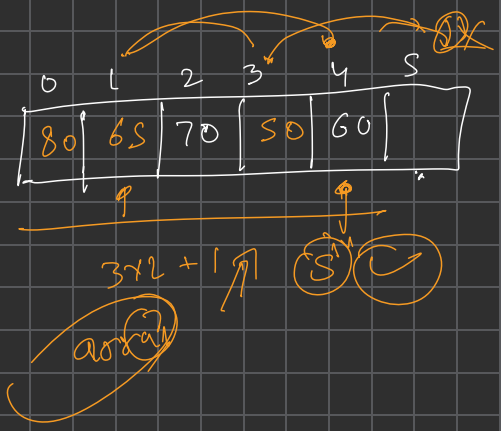
(3) Up element ko correct position

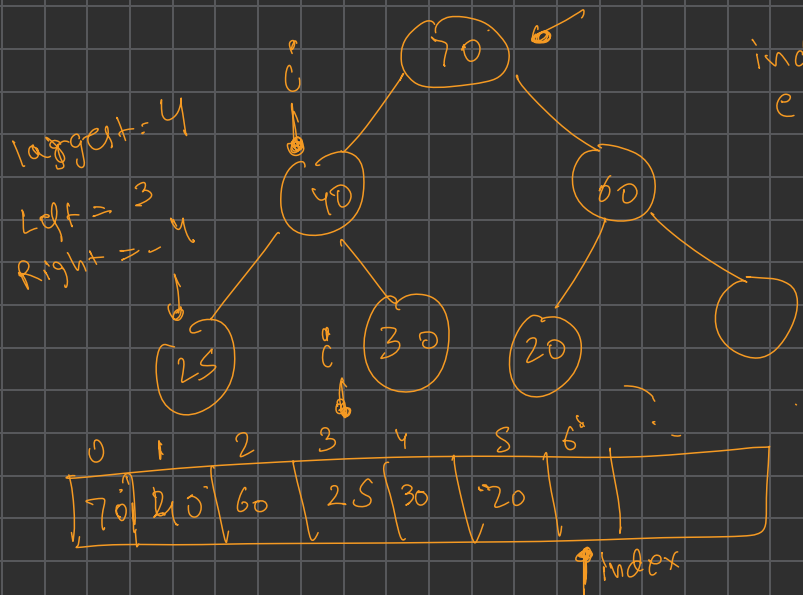


max









index : next element where it will be inw